

Han--Li--Sun	β -	$\beta > 0$	weighted Gaussian density
2			
1. Han--Li--Sun	ε -		
		$\Psi_p(X_0, T; T - r^2) < 1 + \varepsilon_{\text{HLS}},$	(1.1)
	$\varepsilon_{\text{HLS}} > 0$	$\min\{\varepsilon_{\text{HLS}}, 1/2\}$	$0 < \varepsilon_{\text{HLS}} < 1$ (1 +
$\varepsilon_{\text{HLS}}, 2)$	$\Psi_p < 2$	(1.1)	
2. “ ”		2	1
	(1, 2)	1	
3. $\beta > 0$		White	Han--Li--Sun (1.1)

“weighted Gaussian density < 2”

(1.1) $1 + \varepsilon_{\text{HLS}}$

1.

Han--Li--Sun					
		$\partial_t F = \frac{\cos^2 \alpha \, H - \beta \sin^2 \alpha \, V}{\cos^2 \alpha + \beta \sin^2 \alpha},$	$\beta \geq 0,$	(1.2)	
α Kähler	$\beta = 0$	(1.2)	$\partial_t F = H$	$\cos \alpha \geq \delta > 0$	$p \geq p_0(\beta, \delta)$

$$\Psi_p(X_0, t_0; t) = \int_{\Sigma_t} \phi(F) \, \rho_{X_0, t_0}(F, t) \frac{1}{\cos^p \alpha} \, d\mu_t, \tag{1.3}$$

ϕ X_0 1 ρ

“weighted Gaussian density < 2”

- $r > 0$ $\Psi_p(X_0, T; T - r^2) < 2$
-

$$\Theta_p^w(X_0,T):=\lim_{r\downarrow 0}\Psi_p(X_0,T;T-r^2)<2. \tag{1.4}$$

$$\text{Han--Li--Sun} \tag{1.1}$$

2

$$2. \qquad 2$$

$$2 \qquad 0<\cos\alpha\leq 1$$

$$2.1 \quad D_p<2$$

$$\mathcal{C}=\sum_{j=1}^Nm_j|P_j|,\qquad m_j\in\mathbb{N}, \tag{2.1}$$

$$\text{K\"ahler} \qquad c_j=\cos\alpha_{P_j}\in(0,1]$$

$$D_p(\mathcal{C})=\sum_{j=1}^Nm_jc_j^{-p}. \tag{2.2}$$

$$D_p(\mathcal{C})<2 \qquad \sum_jm_j=1 \qquad c \tag{2.3}$$

$$c^{-p}<2 \quad \Longleftrightarrow \quad c>2^{-1/p}.$$

$$c>2^{-1/p} \qquad D_p<2 \qquad c<1 \qquad D_p=c^{-p}>1$$

$$0< c_j\leq 1 \quad p>0 \quad c_j^{-p}\geq 1$$

$$\sum_{j=1}^Nm_j\leq \sum_{j=1}^Nm_jc_j^{-p}=D_p(\mathcal{C})<2. \tag{2.4}$$

$$2 \qquad 1 \tag{2.1} \qquad 1 \qquad (2.3) \quad c^{-p}<2$$

$$D_p=c^{-p}$$

$$2.2 \qquad \text{HLS}$$

$$D_p<2 \qquad (2.2) \qquad D_p=1$$

$$[1,2) \qquad c^{-p} \qquad 0<\varepsilon<1$$

$$1+\varepsilon<c^{-p}<2 \tag{2.5}$$

$$c \in (2^{-1/p}, (1+\varepsilon)^{-1/p}) \qquad D_p < 2 \quad D_p > 1 + \varepsilon \qquad c = 1$$

$$D_p < 1 + \varepsilon_{\mathrm{HLS}}$$

$$c \in (0,1] \qquad \mathbb{C}^2 = \mathbb{R}^4 \qquad e_1, e_2, e_3, e_4 \qquad J e_1 = e_2 \quad J e_3 =$$

$$e_4$$

$$P_c = \operatorname{span} \left\{ e_1, c e_2 + \sqrt{1 - c^2} \, e_3 \right\}. \tag{2.6}$$

Kähler

$$\omega\Big(e_1, c e_2 + \sqrt{1 - c^2} \, e_3\Big) = c. \tag{2.7}$$

$$P_c \quad \text{K\"ahler} \qquad c \qquad P_c \qquad A=H=0 \quad \alpha \qquad V=0 \qquad (1.2)$$

$$~~~~~\text{``}~~~~~2~~~~~\text{HLS}~~~~~\text{''}$$

3.

$$\cos \alpha = *F_t^* \omega \geq \delta > 0, \qquad |A_t|^2 \leq \frac{C}{T-t} \tag{3.1}$$

$$\cos^{-p}\alpha$$

- Kähler
-
- $\Theta_p^w < 2 \qquad \Theta_p^w < 1 + \varepsilon_{\mathrm{HLS}}$
- (1.2) “ ” White
- $|A|^2 \leq C/(T-t)$ Type-I

Han--Li--Sun Theorem 5.7 λ -

$$\text{“holomorphic curve in } \mathbb{C}^2\text{”} \qquad \text{“} \qquad \text{K\"ahler} \qquad \cos \alpha = 1\text{”}$$

$$(1.2)$$

Wang Proposition 5.2

$$\beta > 0$$

4.

4.1 Han--Li--Sun

M Kähler Σ $F:\Sigma\times[0,T)\rightarrow M$ (1.2) $\cos\alpha\geq\delta>0$ $\varepsilon_{\text{HLS}}>0$

Han--Li--Sun Corollary 4.2 X_0 $r_{X_0}>0$

$$\Psi_p(X_0,T;T-r_{X_0}^2)<1+\varepsilon_{\text{HLS}}.$$
 (4.1)

$\varepsilon>0$ F $[0,T+\varepsilon)$

$|A|^2\leq C/(T-t)$

Han--Li--Sun Corollary 4.2 (4.1) (X_0,T)

(X_0,T)

$|A|$ $t\uparrow T$ (x_i,t_i) $t_i\uparrow T$

$|A|(x_i,t_i)\longrightarrow\infty.$ (4.2)

$X_i=F(x_i,t_i)$ M $X_i\rightarrow X_0\in M$ (X_0,T) $r>0$ $K<\infty$

$|A|\leq K$ i (X_i,t_i) (4.2) (X_0,T)

(4.1) Corollary 4.2

$$\sup_{\Sigma\times[0,T)}|A|<\infty.$$
 (4.3)

Han--Li--Sun Theorem 3.1 $\varepsilon>0$ $[0,T+\varepsilon)$

4.2

4.1 (1.4)

$$\Theta_p^w(X_0,T)<1+\varepsilon_{\text{HLS}},$$
 (4.4)

T

(4.4) X_0 $r_{X_0}>0$ (4.1) 4.1

$$\Theta_p^w\leq 1+\varepsilon_{\text{HLS}}$$

5.

2

(2.1)--(2.2)

$c_j = 1$

$\Theta_p^w < 2 \implies \Theta_p^w = 1.$

(5.1)

$\Psi_p < 1 + \varepsilon_{\text{HLS}}$

Corollary 4.2

4.1

“ $c_j = 1$ ”

”

Han--Li--Sun

$\beta > 0$

Kähler

6.

$\beta = 0$

$\beta = 0$

(1.2)

Wang Proposition 5.2

$\cos \alpha \geq \delta > 0$

(3.1)

weighted-density-< 2

“+ White”

2

1

White

< 2

$\beta = 0$

$\beta > 0$

7.

$\beta > 0$

$F : \Sigma \times [0, T) \rightarrow M^4$

Kähler

β -

$\cos \alpha \geq \delta > 0$

$p \geq p_0(\beta, \delta)$

(X_0, T)

$r_{X_0} > 0$

$\Psi_p(X_0, T; T - r_{X_0}^2) < 1 + \varepsilon_{\text{HLS}},$

$\varepsilon > 0$

$[0, T + \varepsilon)$

“2”

$\beta > 0$

[1]

X. Han, J. Li and J. Sun, *Gradient Flow for β -Symplectic Critical Surfaces*, Ann. Inst. H. Poincaré C Anal. Non Linéaire 41 (2024), 1083--1116

Theorem 3.1

Corollary 4.2

Theorem 5.7

[2]

M.-T. Wang, *Mean Curvature Flow of Surfaces in Einstein Four-Manifolds*, J. Differential Geom. 57 (2001), 301--338

Proposition 5.2

[3]

B. White, *A Local Regularity Theorem for Mean Curvature Flow*, Ann. of Math. 161 (2005), 1487--1519

[4] J. Chen and J. Li, *Singularities of Codimension Two Mean Curvature Flow of Symplectic Surfaces*,
arXiv:math/0208227 Theorem 5.1